Question	Answer	Marks	Partial Marks
13	45[.0] or 44.99 to 45.00	**2**	**M1** for $\frac{1}{2}\times 13\times 11\times \sin 39$ oe
14	49 000	**3**	**M1** for $4.9\times(10\,000\,000)^2$ **M1** for $\div(100\,000)^2$ OR **M1** for 1 cm : 100 km **M1** for $4.9\times(\textit{their}\ 100)^2$ OR **M2** for $(\sqrt{4.9}\times 10\,000\,000\div 100\,000)^2$ or **M1** for $\sqrt{4.9}\times 10\,000\,000\div 100\,000$
15	128	**3**	**M1** for $y=\frac{k}{x^2}$ **M1** for $y=\frac{\textit{their}\ k}{\left(\frac{1}{2}\right)^2}$ OR **M2** for $\frac{2\times 4^2}{\left(\frac{1}{2}\right)^2}$ or **M1** for $2\times 4^2=y\times\left(\frac{1}{2}\right)^2$
16	109.3 or 109.26 to 109.27	**3**	**M2** for $\frac{12\sin 39}{8}$ or **M1** for $\frac{8}{\sin 39}=\frac{12}{\sin(...)}$ oe
17	6.28 or 6.283 to 6.284	**3**	**M2** for $\frac{45}{360}\times\pi\times 5^2$ oe **and** $\frac{45}{360}\times\pi\times 3^2$ oe or **M1** for $\frac{45}{360}\times\pi\times 5^2$ oe or $\frac{45}{360}\times\pi\times 3^2$ oe or $\pi\times 5^2-\pi\times 3^2$ oe
18	$\frac{x^2-3x-8}{2(x+1)}$ or $\frac{x^2-3x-8}{2x+2}$ final answer	**3**	**B1** for common denominator $2(x+1)$ or $2x+2$ **M1** for $x(x+1)-2(2x+4)$ or better
19(a)	$\begin{pmatrix}19 & 22\\ 43 & 50\end{pmatrix}$	**2**	**B1** for 2 or 3 elements correct

Question	Answer	Marks	Partial Marks
19(b)	–2 final answer	**1**	
20	$\frac{147}{160}$ oe	**3**	**M2** for $\frac{1}{10}\times\frac{3}{4}+\frac{9}{10}\times\frac{15}{16}$ or **M1** for $\frac{1}{10}\times\frac{3}{4}$ or $\frac{9}{10}\times\frac{15}{16}$
21(a)	Translation $\begin{pmatrix}-1\\-5\end{pmatrix}$	**2**	**B1** for each
21(b)	Correct reflection at (6, 2), (6, 6), (7, 6), (7, 3)	**2**	**B1** for three correct vertices
22	2592	**4**	**M3** for $1.2\times100\times60\times60\times6\div1000$ oe or **M2** for $1.2\times60\times60\times6$ oe or **M1** for figs $12\times$ figs 6 or 60×60 or correct conversion e.g. their value in $\text{cm}^3\div1000$ their value in $\text{m}^3\times1000$ 1.2×100 $6\div10\,000$
23	2, 5 3 $A.....B'$ $\subset$	**4**	**B1** for each
24(a)	19	**2**	**M1** for $3(2^x)-5$ soi or for f(8)
24(b)	$\frac{x+5}{3}$ oe final answer	**2**	**M1** for correct first step $y+5=3x$ or $\frac{y}{3}=x-\frac{5}{3}$ or $x=3y-5$
25(a)	$-\frac{1}{3}\mathbf{q}+\frac{1}{2}\mathbf{p}$ oe	**2**	**M1** for correct unsimplified answer or correct route
25(b)	$\frac{1}{2}\mathbf{p}+\frac{1}{2}\mathbf{q}$ oe	**2**	**M1** for correct unsimplified answer or correct route

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